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Title: Emergency letter for Hyperammonemia- Cause yet Unknown

Introduction

Ammonia is an important source of nitrogen and is required for amino acid synthesis. It is also necessary for normal acid-base balance. When present in high concentrations, ammonia is toxic. Endogenous ammonia intoxication can occur when there is impaired capacity of the body to excrete nitrogenous waste, as seen with congenital enzymatic deficiencies for example urea cycle disorders or a number of other disorders that perturb the liver can result in *hyperammonemia* and mimic the effects of a urea cycle disorder. It might be secondary to deficiencies in the urea cycle substrate which is the case in organic acidemia and fatty acid oxidation disorders. A variety of environmental causes and medications may also lead to ammonia toxicity. Hyperammonemia refers to a clinical condition associated with elevated ammonia levels manifested by a variety of symptoms and signs, including significant central nervous system (CNS) abnormalities including seizure, encephalopathy, poor feeding, nusease, vomiting and abnormal pattern of breathing

Interpretation

The interpretation of plasma ammonia values can be difficult and should always be done in conjunction with the history. Normal values should be less than 50 $\mu\text{mol/l}$ but mildly raised values are common (up to 80 $\mu\text{mol/l}$). Artifactually high values can be caused by muscle activity, haemolysis, or delay in separating the sample. Capillary samples are often haemolysed or contaminated and therefore should be not be used. Values up to 80 $\mu\text{mol/l}$ are common in patients with urea cycle disorders, even those with good metabolic control. Plasma ammonia concentrations are usually $>100 \mu\text{mol/l}$ during an episode of decompensation but, on rare occasions, they may not be raised in the early stages.

In neonates, any illness may be responsible for values up to 180 $\mu\text{mol/l}$. In any patient, values above**200 $\mu\text{mol/l}$** require urgent attention. In all cases the history is important. The symptoms, signs, and treatment may all provide clues. There is no simple test that will indicatea diagnosis, such as blood gases.

Investigations and management should be very urgent

Investigations

- TMS (Acylcarnitine and plasma amino acids screen):** can be collected and stored in our hospital lab even in week/ holidays and after the hours Top
- Repeat ammonia immediately if $>70 \mu\text{mol/l}$ in children, 100 $\mu\text{mol/l}$ in neonates

- Lactate
- Plasma amino acids (this should be done urgently if there is significant hyperammonaemia and a urea cycle disorder is suspected)
- Blood spots in Guthrie card for acyl carnitine profile
- Urine organic acids
- Urine orotic acid
- DNA extraction and storage
- Plasma urea & electrolytes, liver function tests, clotting studies
- Blood culture if sepsis is suspected

Management: Management is very urgent- Do not delay.

- Contact metabolic center/metabolic consultant for management advice and potential patient transfer to tertiary hospital
- If the repeat ammonia $> 250 \mu\text{mol/l}$, arrange for urgent transfer to specialist center, but continue urgent management as you are arranging the transfer
- Careful and frequent clinical assessment including GCS and pupils even if the patient does not appear encephalopathic (baseline if the child deteriorates at transfer)
- Note and document any degree of encephalopathy
- The results of tests are urgent so call the lab and discuss them
- Keep NPO and start intravenous IVF till the patient is stable and diagnosis is known
- Give dextrose 200 mg/kg once (2 ml/kg of D10%) if there is hypoglycemia
- Give normal saline 10 ml/kg unless the peripheral circulation is poor or the patient is frankly shocked, then give 20 ml/kg normal saline as a bolus immediately after the glucose. Repeat the saline bolus if the poor circulation persists as for a shocked non-metabolic patient.
- Continue with D10% at 5 ml/kg/h until next solution ready.
- Put the patient on IVF at rate of 1.5 maintenance of dextrose 10%+0.45 NaCl, providing GIR of 6-8 mg/kg/min
- It is assumed that the patient will be given sodium benzoate and sodium phenylbutyrate at full dose therefore use D10%. If not giving full doses, use 0.18% Saline plus D10%
- **Medications:** The medicines should be given as soon as possible as continuous intravenous infusions. Sodium benzoate, sodium phenylbutyrate and arginine can be given together (the maximum concentration for infusion being no more than 2.5 gram of each drug to 50ml of D5% or D10%.
- In an emergency, a loading dose should be given initially followed by the maintenance dose.

Drug	Loading dose over 90 minutes	Then maintenance dose over 24 hours	Sodium content of daily maintenance
Sodium benzoate	250 mg/kg	250 mg/kg	3.5 mmol/kg/day
Sodium phenylbutyrate	250 mg/kg	250 mg/kg	2.8 mmol/kg/day
Arginine	200 mg/kg	200 mg/kg	Nil

- After the initial treatment, it is strongly recommended that the doses are discussed with the regional metabolic center.
- Consider giving a single oral dose of N-carbamyl glutamate (250 mg/kg). This is the treatment of choice for N-acetylglutamate synthase deficiency (very rare). It may also be of benefit for hyperammonaemia secondary to organic acidurias.
- **When to consider dialysis.** As the risk for irreversible cerebral damage is directly related to NH_3 concentration in plasma, immediate measures to decrease plasma levels of NH_3 are necessary in patients with high or rapidly rising concentrations. There is no evidence based data on the precise levels which should lead to the invasive procedure of extracorporeal detoxification, but levels $>250 \mu\text{mol/l}$ in neonates or young infants should lead to preparation for extracorporeal detoxification, and if levels are $>500 \mu\text{mol/l}$ this procedure should be started without delay.
- Treat any infection and constipation (which increases ammonia absorption from the gut).

local diabetic protocol rather than reducing the glucose intake

- Potassium. Hypokalaemia is common so plasma potassium concentration should be monitored carefully. Potassium should be added once urine flow is normal and the plasma potassium concentration is known.

Monitor the patient

- If there is any hint of incipient encephalopathy (lethargy, unusual behavior, etc.) start neurological observations at least hourly and seek specialist help. Under these circumstances, fluid volumes should be reduced and given via central line as concentrated solutions to minimize risk of cerebral edema.
- Reassess after 4-6 hours or earlier.
- Clinical assessment should include Glasgow coma score and vital signs.
- Blood tests:
 - Blood pH and gases
 - Ammonia
 - Urea & electrolytes
- If improving, continue intravenous fluids and medicines (see the previous section)
- If deteriorating (clinical state, hyperammonaemia), seek specialist help. Dialysis may need to be considered urgently. Note peritoneal dialysis is less efficient. Exchange transfusion is dangerous and should not be used.

Important note

- in case the patient presented with hyperaamonaemia and found to have sever persistent metabolic acidosis with high anion gab

after collection of basic metabolic workup including acylacarnitine following medications c should be used

Carnitine (iv with initial dose of 100 mg/kg stat then
to be giving as 50mg/kg/day
Oral biotin 10 mg once a day

Please update medical genetic on call with patient condition.

Reference:

Modified from BIMDG protocol[1]

Suggested guidelines for the diagnosis and management of urea cycle disorders.

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